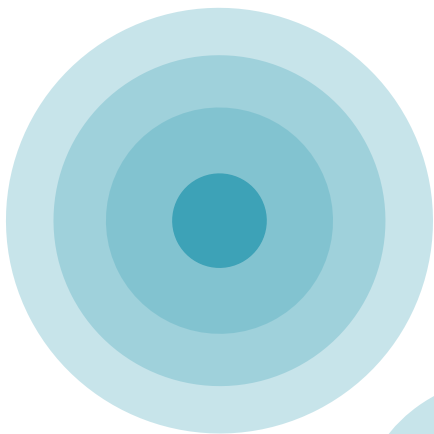
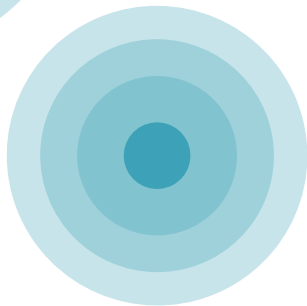
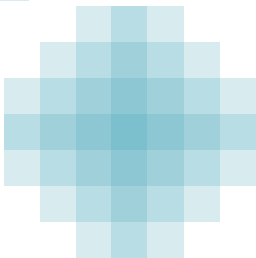
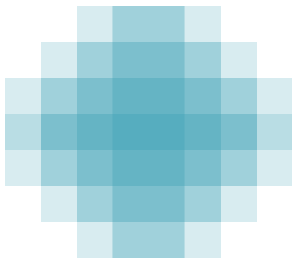


Parametric Distribution

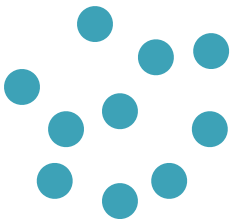


α_{θ}





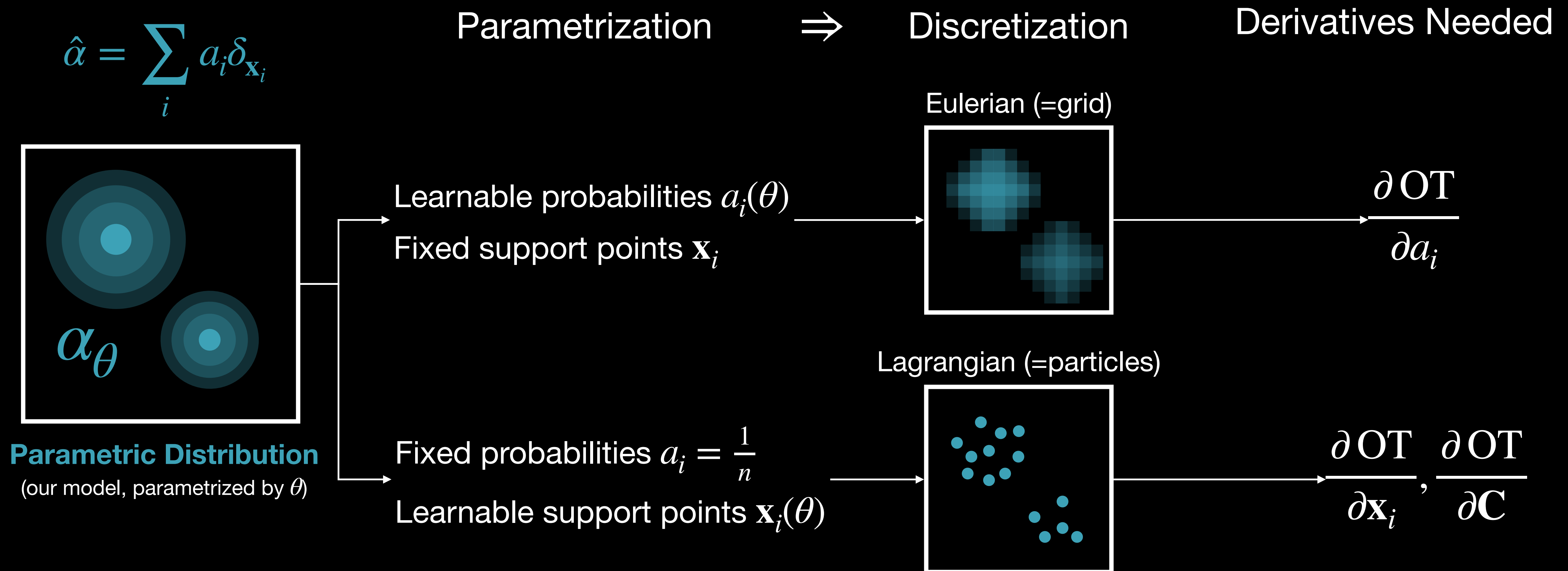
estimate OT loss from finite samples



$$\hat{\alpha} = \sum_i a_i \delta_{\mathbf{x}_i}$$

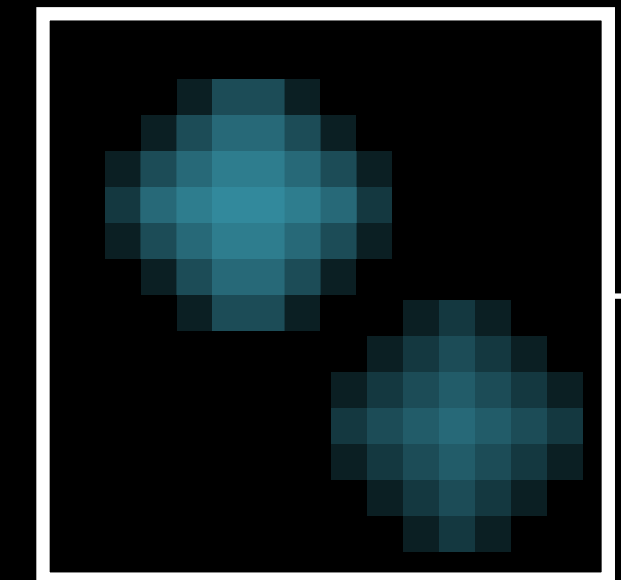
Parametrizations and Derivatives

In most interesting applications, no closed-form OT solution between α_θ, β
 $\Rightarrow$ need to **estimate OT loss from finite samples** (i.e., empirical measures)



Derivatives w.r.t. Histograms

Eulerian (=grid)



$$\frac{\partial OT}{\partial a_i}$$